Three-Pillar Resolution of the Blood-Brain Barrier (BBB) Pharmacokinetic Paradox

Reconciling Low Micromolar In Vivo Circulating SCFAs (1–10 μM) with High Millimolar In Vitro HDAC IC_{50} (500–2,000 μM)

PILLAR 1

BAM Vascular Sensor Relay

Perivascular Border Interface



- **Anatomical Localization:** BAMs populate leptomeningeal, dural, and Virchow-Robin perivascular spaces directly.
- **High MCT1 Expression:** Positioned blood-facing; sense micromolar SCFAs without requiring full parenchymal entry.
- **Secondary Paracrine Relay:** Transduce cues via TGF-β1 & IL-10 across glia limitans to calibrate resting microglial tone.

PILLAR 2

Central Acetate / ACSS2

Nuclear HAT Epigenetic Flux



- **High-Capacity BBB Transport:** Unlike butyrate/propionate, acetate crosses the BBB with significant net flux via endothelial MCT1.
- **Microglial ACSS2 Capture:** Microglia express high ACSS2; directly captures acetate to regenerate intranuclear acetyl-CoA.
- **HAT Substrate Replenishment:** Fuels histone acetyltransferases directly, sustaining chromatin poising without millimolar HDAC IC_{50}.

PILLAR 3

Enteric-Vagal Reflex

Neuro-Humoral Circuit



- **Mucosal Enteric Sensing:** Intestinal luminal SCFAs stimulate enteroendocrine L-cells and subepithelial vagal nerve terminals.
- **Afferent Vagal Conduit:** Signals ascend via nodose ganglion to the nucleus tractus solitarius (NTS) in the dorsal vagal complex.
- **Neuro-Immune Calibration:** Ascending projections modulate parenchymal microglia without requiring direct BBB crossing.

Resolution of the In Vivo Pharmacokinetic Paradox:

Direct in vitro competitive HDAC inhibition requires millimolar concentrations (IC_{50} 500–2,000 μM), but circulating gut SCFAs are only 1–10 μM in vivo. Microglial homeostasis is sustained not by massive parenchymal but